Customer Retention and Churn Risk Analysis

Predictive analytics and customer risk assessment

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1. Executive Summary

Customer attrition represents a direct risk to recurring revenue and customer lifetime value. This analysis evaluates customer behaviour and develops a predictive approach to identify customers at elevated risk of churn, with the aim of supporting more targeted and effective retention activity.

The analysis examined customer usage, charges, service interactions, product adoption, contract status and account characteristics. Exploratory analysis identified several notable relationships with churn, particularly daytime usage, customer-service interactions, monthly charges, contract renewal and data-plan status. Customers without a data plan recorded an observed churn rate of 16.72%, compared with 8.68% among customers with a data plan, while contract renewal showed an especially strong relationship with customer retention.

Logistic Regression, Random Forest and XGBoost were evaluated using accuracy, precision, recall, F1-score and ROC-AUC. Random Forest was selected as the final model, providing the strongest overall balance of predictive performance with 92.50% accuracy, 82.19% precision, 61.86% recall, 70.59% F1-score and 86.32% ROC-AUC. Although XGBoost achieved marginally higher recall, Random Forest performed better across the other principal metrics.

Model interpretation identified DayMins, MonthlyCharge and CustServCalls as the three most influential features within the Random Forest model. Logistic Regression odds ratios and SHAP analysis provided additional insight into the direction and interaction of important predictors, reinforcing that churn risk is influenced by multiple customer characteristics rather than a single factor.

The model was subsequently used to assess predicted churn risk among existing customers and identify potentially actionable customer groups. Customers without contract renewal represented the highest-risk segment, with an average predicted churn probability of 13.86%. High-usage customers without a data plan represented a second notable segment, with an average predicted churn probability of 9.31%. Frequent customer-service interactions were also more prevalent among high-risk customers, although the broader group of customers with two or more service calls had a lower average predicted risk of 4.19% and should therefore be treated as a supporting risk indicator rather than a standalone segment.

The findings support a targeted, risk-based retention strategy rather than a broad approach applied uniformly across the customer base. Priority should be given to proactive contract-renewal interventions, targeted product or plan recommendations for high-usage customers without data plans, and service recovery for customers where repeated service interactions coincide with elevated churn risk.

2. Business Context and Objectives

2.1 Business Context

Customer attrition represents a direct commercial risk for telecom providers, as customers who move to competitors result in lost recurring revenue and reduced customer lifetime value.

The objective of this analysis is to use customer behaviour, usage patterns and plan characteristics to understand the factors associated with churn and identify customers who may be at elevated risk of leaving. The analysis is intended to support a more proactive and targeted approach to customer retention.

2.2 Objectives

The analysis has four primary objectives:

1. Identify key factors associated with customer churn.
2. Develop a model to estimate individual customer churn probability and identify high-risk customers.
3. Identify customer groups exhibiting elevated churn risk.
4. Translate the findings into actionable retention strategies that can improve customer experience and reduce revenue loss.

2.3 Analytical Questions

To address these objectives, the analysis focuses on four questions:

- Which customer characteristics are most strongly associated with churn?
- Which machine-learning approach provides the most effective churn-risk predictions?
- What characteristics are shared by customers with elevated predicted churn risk?
- How can these insights be translated into targeted retention actions?

3. Customer & Churn Overview

3.1 Dataset Overview

The analysis uses customer-level telecom data containing 3,333 customer records and 10 explanatory variables, with customer churn represented by the binary target variable Churn.

The available variables capture several aspects of the customer relationship, including:

- Usage: DayMins, DayCalls, RoamMins, DataUsage

- Charges: MonthlyCharge, OverageFee
- Customer engagement: CustServCalls
- Product relationship: DataPlan
- Contract status: ContractRenewal
- Account tenure: AccountWeeks

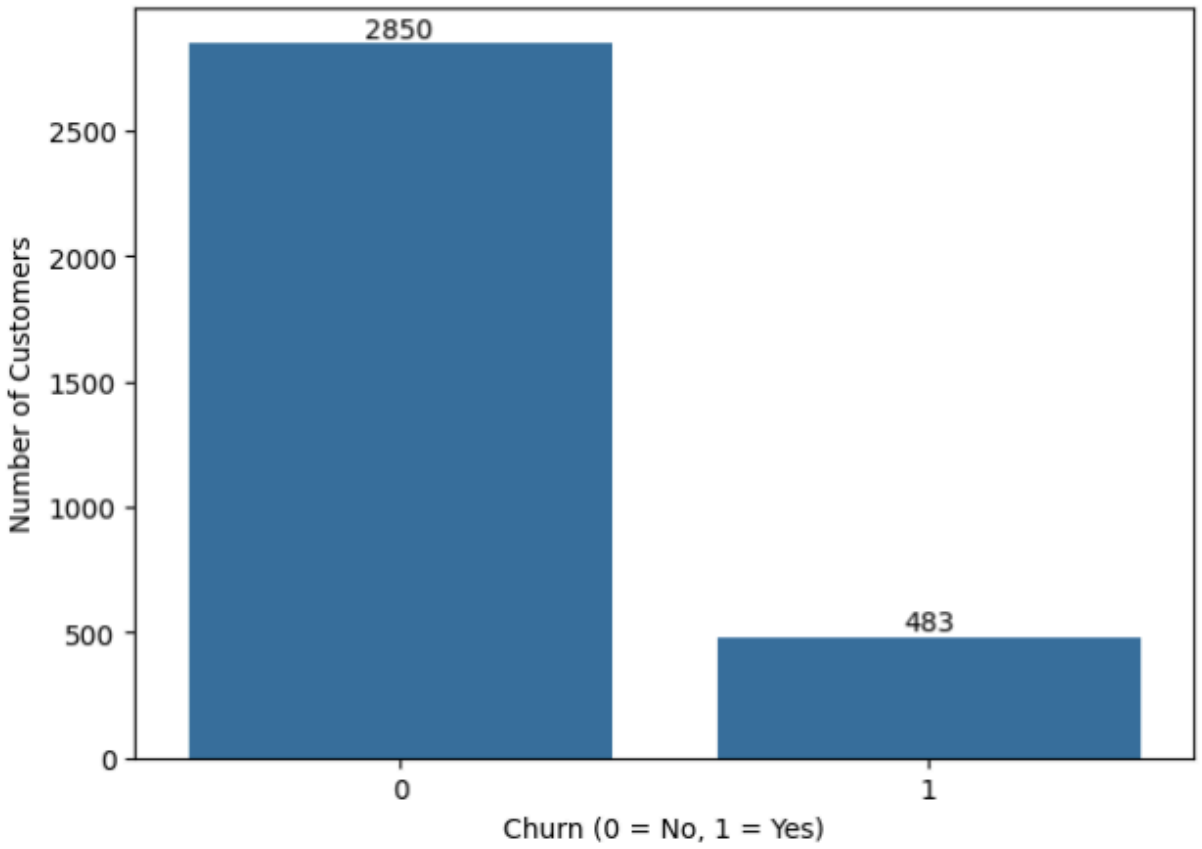
The target variable, Churn, indicates whether a customer has left the service (1) or remained with the company (0).

3.2 Churn Distribution

The dataset contains both churned and retained customers, providing the basis for a binary classification problem.

The analysis identified approximately 14.5% of customers as churners, with the remaining customers classified as non-churners. This represents a moderately imbalanced target, making metrics such as precision, recall, F1-score and ROC-AUC important alongside accuracy when evaluating the predictive models.

Figure 1: Customer Churn Distribution



3.3 Data Quality

Initial inspection found no material data-cleaning requirements. The variables were already in a suitable format for analysis and modelling, and the binary variables were appropriately represented as 0/1.

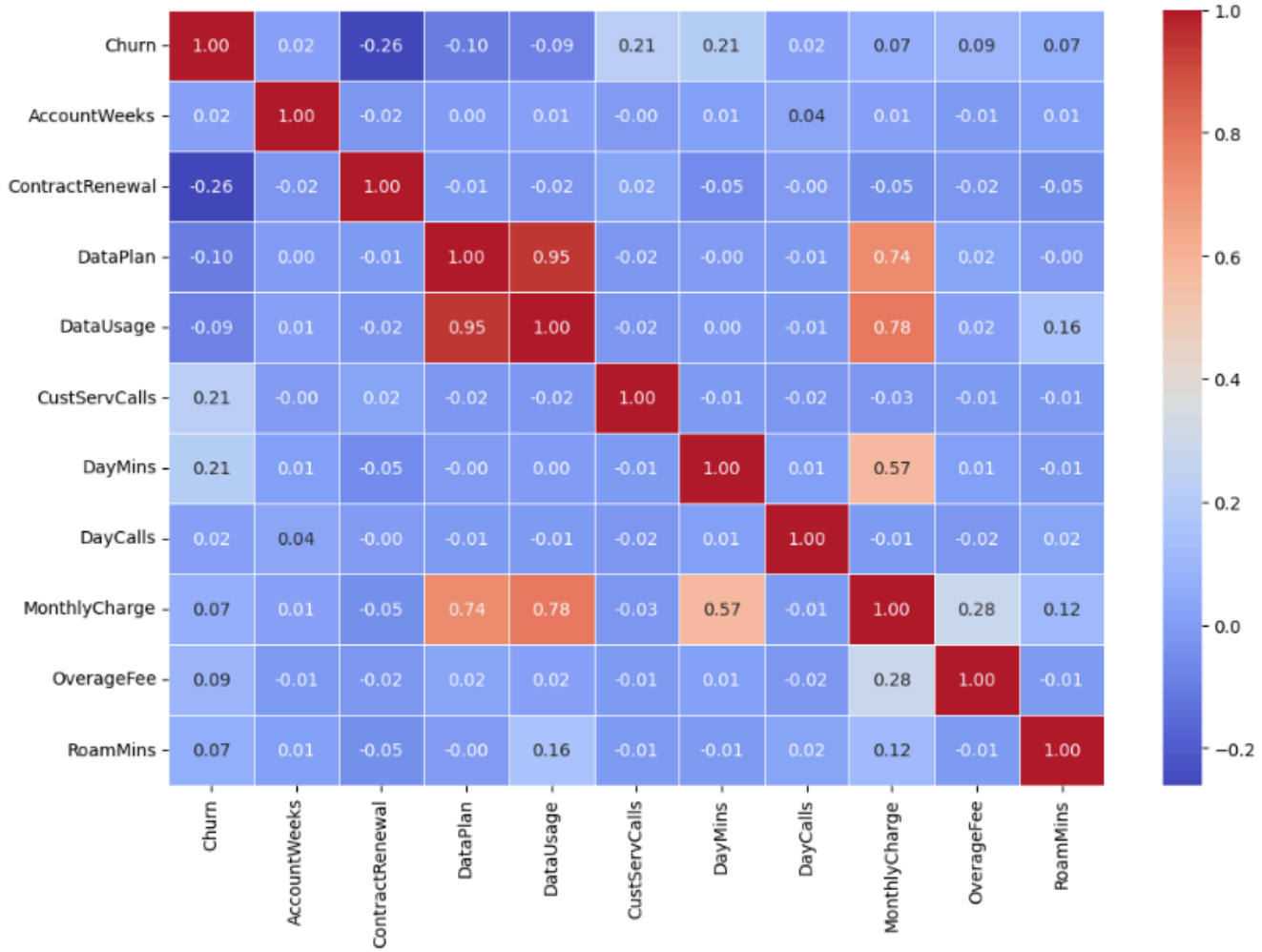
No additional feature engineering was required because the available variables provided suitable representations of customer usage, charges, engagement and account characteristics.

4. Key Drivers of Churn

4.1 Correlation Analysis

The correlation analysis identified several variables with notable associations with customer churn. In particular, DayMins, CustServCalls and MonthlyCharge showed positive relationships with churn, while ContractRenewal and DataPlan showed negative relationships.

Figure 2: Correlation Matrix

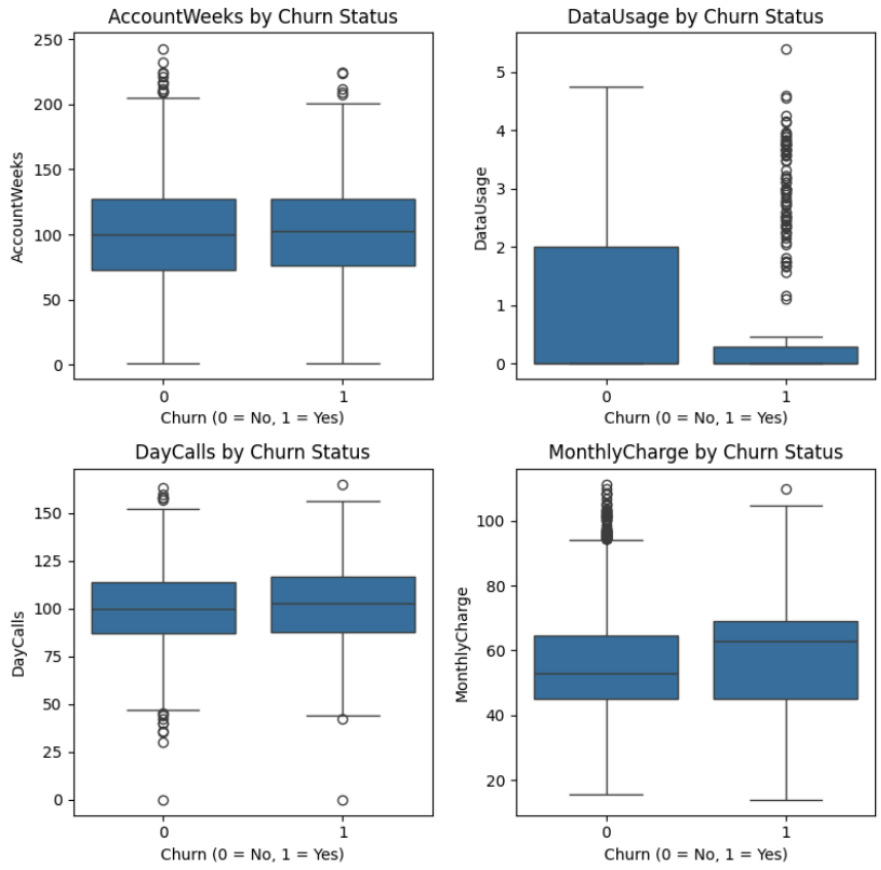


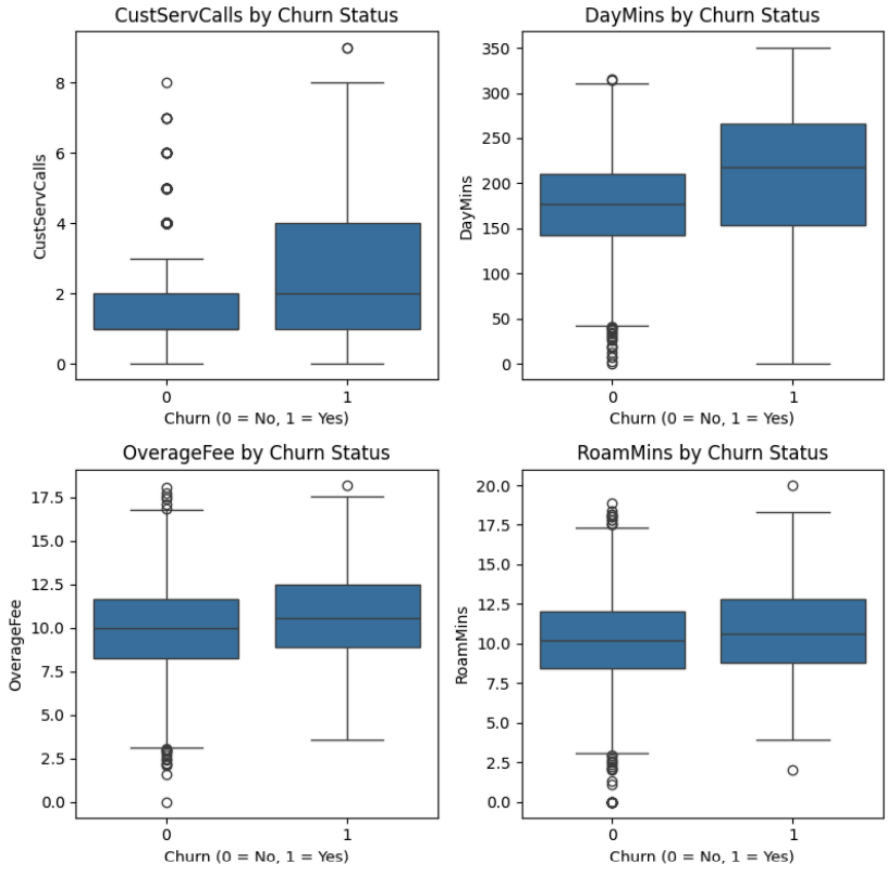
The correlations provide an initial indication of which customer characteristics warrant further investigation. However, correlation alone does not establish causation and does not account for interactions between variables.

4.2 Distribution of Churn Across Key Features

Box plots were used to compare the distributions of key numerical variables between customers who churned and those who remained.

Figure 3: Distribution of Key Numerical Features by Churn Status





The analysis indicates that churned customers tend to exhibit:

- Higher daytime usage (DayMins)
- More customer-service interactions (CustServCalls)
- Higher monthly charges (MonthlyCharge)

These observations are consistent with the subsequent predictive modelling, where these variables were also identified as important predictors.

4.3 Contract Renewal and Data Plan

The categorical analysis provided two particularly clear differences in observed churn behaviour.

Customers without a **data plan** had an observed churn rate of approximately 16.72%, compared with 8.68% among customers with a data plan.

Contract renewal showed an even stronger relationship with churn, with customers without renewal displaying substantially higher churn than customers with renewal.

Figure 4: XGBoost Feature Importance

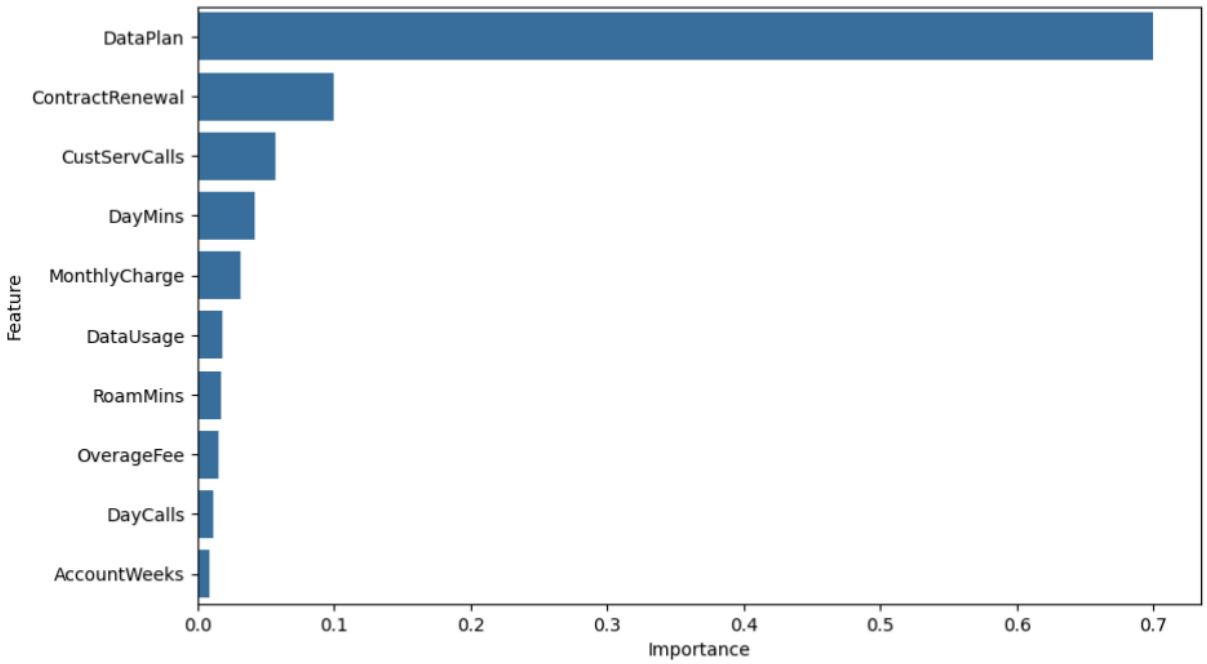


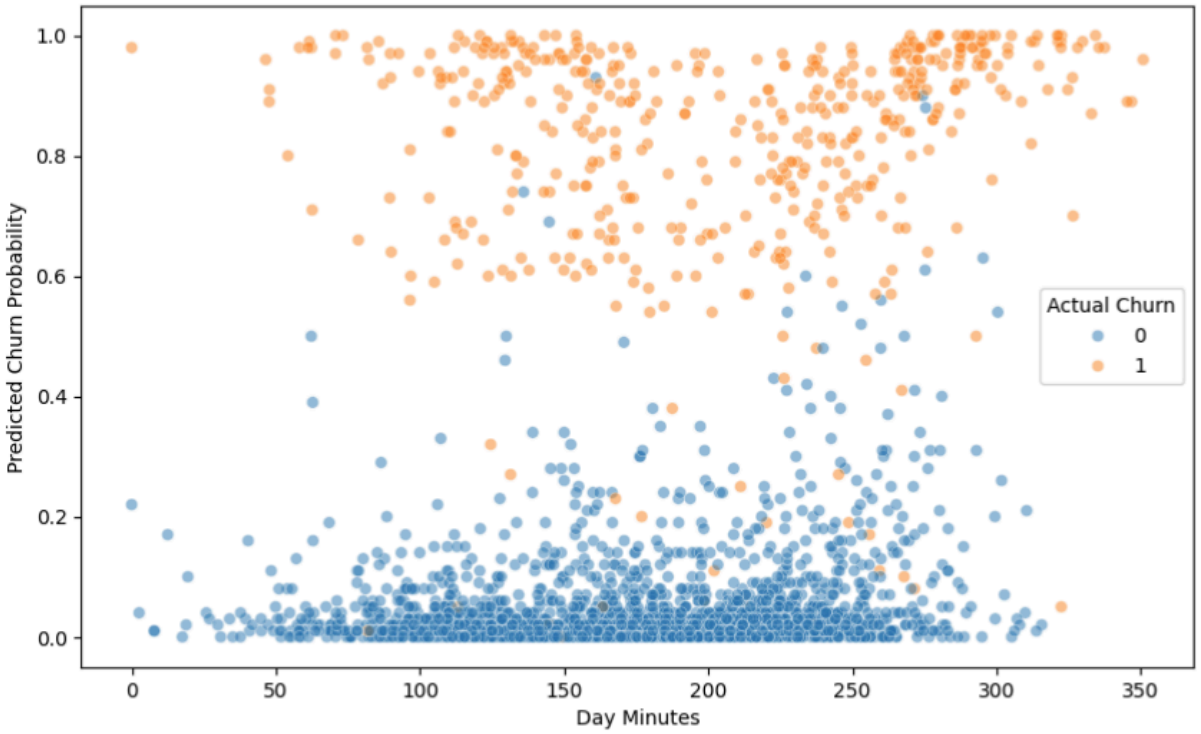
Table 1: Data Plan Customers and Churn Rate

Data Plan	Customers	Churn Rate
No	2411	16.72%
Yes	922	8.68%

4.4 Usage and Predicted Churn

The relationship between customer usage and modelled churn risk was further examined using DayMins and predicted churn probability.

Figure 5: Day Minutes vs Predicted Churn Probability



The plot shows that higher daytime usage is generally associated with a greater concentration of customers receiving elevated predicted churn probabilities. However, customers with similar usage levels can receive substantially different risk predictions, demonstrating that churn risk is influenced by multiple customer characteristics rather than usage alone.

4.5 Key Findings from EDA

The EDA provides four primary observations:

- Higher daytime usage is associated with increased churn risk.
- More customer-service interactions are associated with increased churn risk.
- Customers without a data plan exhibit higher observed churn.
- Contract renewal is strongly associated with lower churn.

These findings establish the initial hypotheses that are subsequently evaluated using the predictive models.

5. Data Preparation

5.1 Data Quality Assessment

An initial assessment of the dataset was performed to identify missing values, inconsistent data types and other potential data-quality issues.

No material data-cleaning requirements were identified, and the available variables were suitable for subsequent analysis and modelling.

5.2 Feature Engineering

No additional feature engineering was performed.

The existing variables already provided meaningful representations of customer usage, charges, account characteristics, product adoption and contract status. Introducing additional derived variables was therefore not considered necessary for the initial modelling exercise.

5.3 Feature Scaling

Numerical features were standardised using StandardScaler to ensure that variables measured on different scales could be used consistently by scale-sensitive algorithms, particularly Logistic Regression.

5.4 Train-Test Split

The dataset was divided into training and testing subsets. The training data was used to fit the predictive models, while the held-out test data was reserved for evaluating their performance on previously unseen customers.

This separation ensures that model performance reflects its ability to generalise beyond the observations used during training.

6. Predictive Modelling

6.1 Modelling Approach

Three classification algorithms were evaluated to predict customer churn:

- Logistic Regression — provides an interpretable baseline and allows the relationship between individual features and churn odds to be examined.
- Random Forest — captures non-linear relationships and interactions between customer characteristics.
- XGBoost — a gradient-boosting approach capable of modelling complex relationships and providing an additional benchmark against the Random Forest.

All models were trained using the prepared training dataset and evaluated on the held-out test dataset.

6.2 Evaluation Metrics

Model performance was assessed using:

- Accuracy — proportion of all predictions that were correct.
- Precision — proportion of customers predicted to churn who actually churned.
- Recall — proportion of actual churners successfully identified by the model.
- F1-score — harmonic mean of precision and recall.
- ROC-AUC — measures the model's ability to distinguish between churners and non-churners across classification thresholds.

Because the objective is to identify customers at risk of leaving, precision and recall are particularly relevant alongside overall predictive performance.

6.3 Logistic Regression

Logistic Regression was established as an interpretable baseline model for the churn prediction task.

The model achieved the following performance.

Table 2: Logistic Regression Performance

Metric	**Performance**
Accuracy	85.76%
Precision	52.63%
Recall	20.62%
F1-score	29.63%
ROC-AUC	**80.91%**

Figure 6: Logistic Regression Confusion Matrix

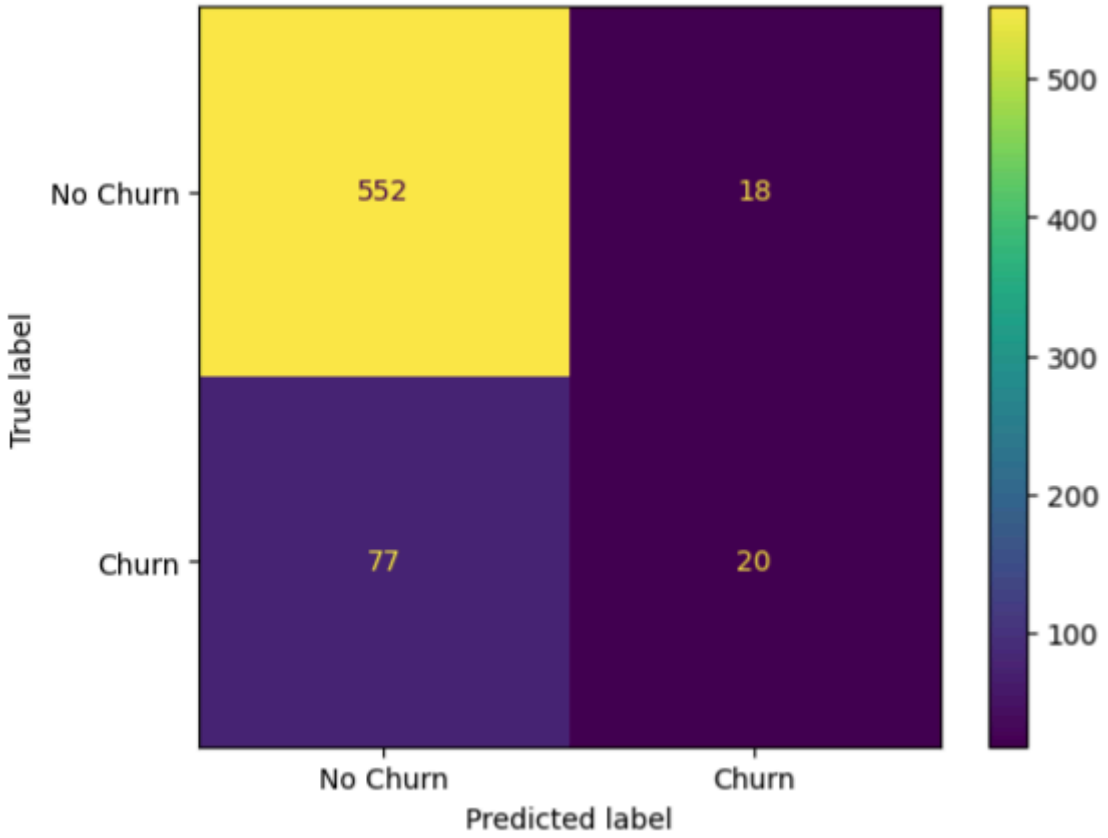
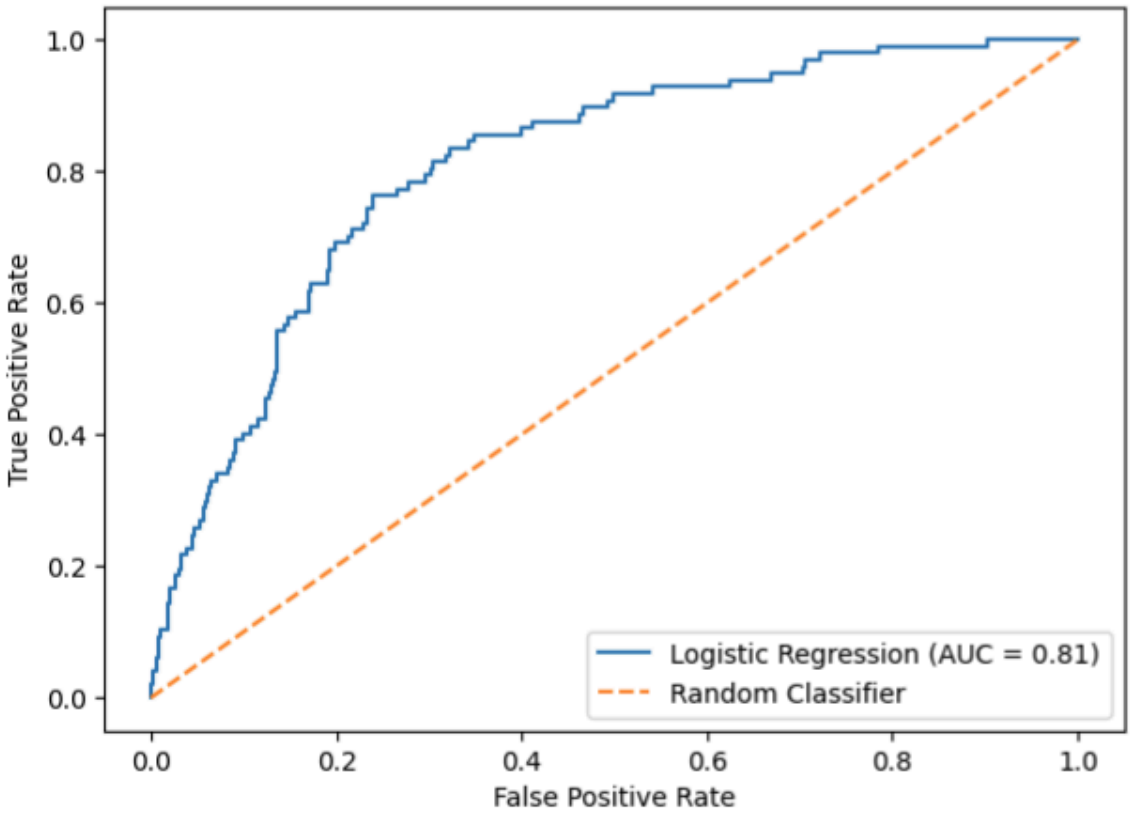


Figure 7: ROC Curve - Logistic Regression



The model provides useful interpretability through its coefficients and odds ratios, which are examined later in the report.

6.4 Random Forest

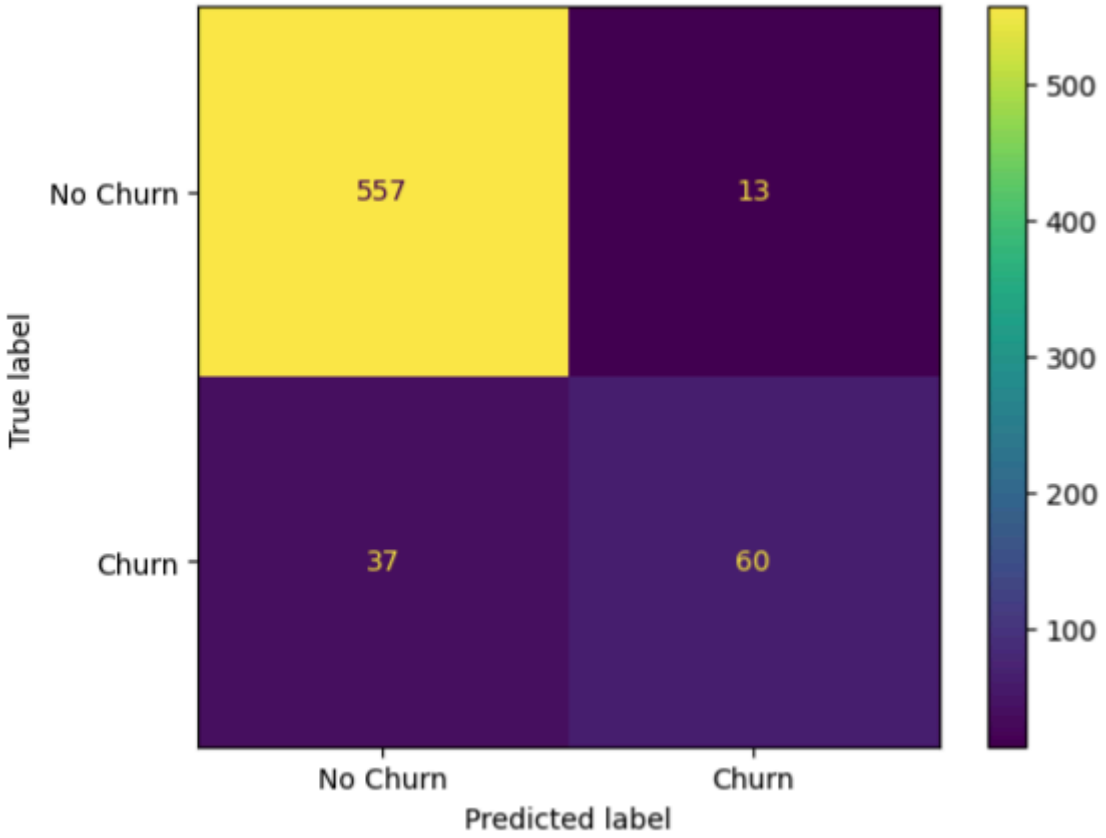
Random Forest was evaluated to capture potentially non-linear relationships between customer characteristics and churn.

The model achieved the following performance.

Table 3: Random Forest Performance Metrics

Metric	Performance
Accuracy	92.50%
Precision	82.19%
Recall	61.86%
F1-score	70.59%
ROC-AUC	86.32%

Figure 8: Random Forest Confusion Matrix



The substantial improvement over Logistic Regression across the majority of evaluation metrics indicates that the relationships within the dataset are not adequately represented by a purely linear classification approach.

6.5 XGBoost

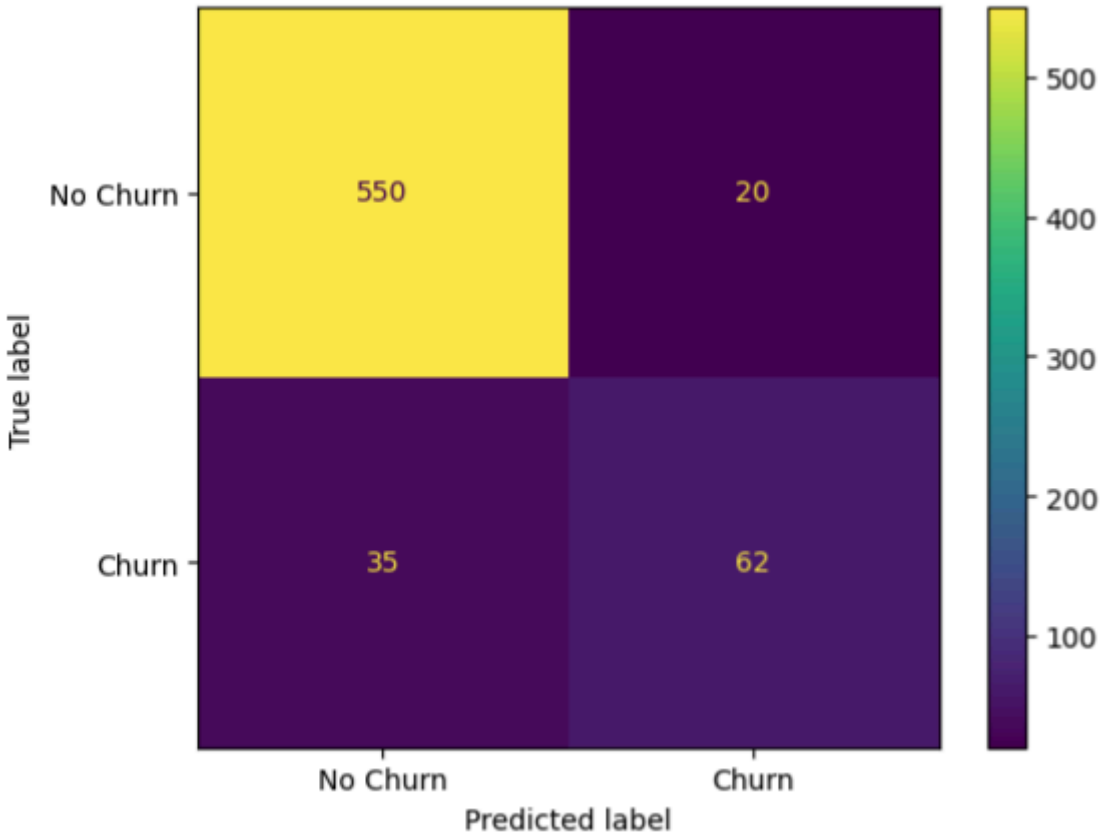
XGBoost was evaluated as a further tree-based modelling approach.

The model achieved the following performance.

Table 4: XGBoost Performance Metrics

Metric	**Performance**
Accuracy	91.75%
Precision	75.61%
Recall	63.92%
F1-score	69.27%
ROC-AUC	82.49%

Figure 9: XGBoost Confusion Matrix



XGBoost achieved marginally higher recall than Random Forest, but performed below Random Forest across the other principal evaluation metrics.

7. Model Performance & Selection

7.1 Comparative Performance

The majority-class baseline achieved 85.46% accuracy by predicting the non-churn class for every customer in the test set. This establishes the performance level that can be achieved without learning any relationship between the customer characteristics and churn.

Logistic Regression achieved 85.76% accuracy, representing only a 0.30% improvement over the baseline. However, its ROC-AUC of 80.91% and precision of 52.63% indicate that it provides meaningful predictions despite the limited improvement in overall accuracy.

Both tree-based models exceeded the baseline. Random Forest achieved 92.50% accuracy, a 7.04% improvement over the baseline, while XGBoost achieved 91.75%, a 6.29% improvement. Both models also demonstrated stronger recall and F1-scores than Logistic Regression, indicating improved ability to identify customers who churn.

Table 5: Comparative Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Majority-Class Baseline	85.46%	—	—	—	—
Logistic Regression	85.76%	52.63%	20.62%	29.63%	80.91%
Random Forest	92.50%	82.19%	61.86%	70.59%	86.32%
XGBoost	91.75%	75.61%	63.92%	69.27%	82.49%

7.2 Model Selection

Random Forest was selected as the final model. It achieved the strongest overall performance across the evaluation metrics, with 92.50% accuracy, 82.19% precision, 70.59% F1-score and 86.32% ROC-AUC.

XGBoost achieved a marginally higher recall of 63.92%, compared with 61.86% for Random Forest. However, this 2.06 percentage-point improvement in recall was accompanied by lower precision, F1-score, accuracy and ROC-AUC.

Given the objective of identifying customers at risk of churn while avoiding excessive false-positive retention interventions, Random Forest provides the stronger overall balance between identifying churners and maintaining prediction precision.

Final Model: Random Forest

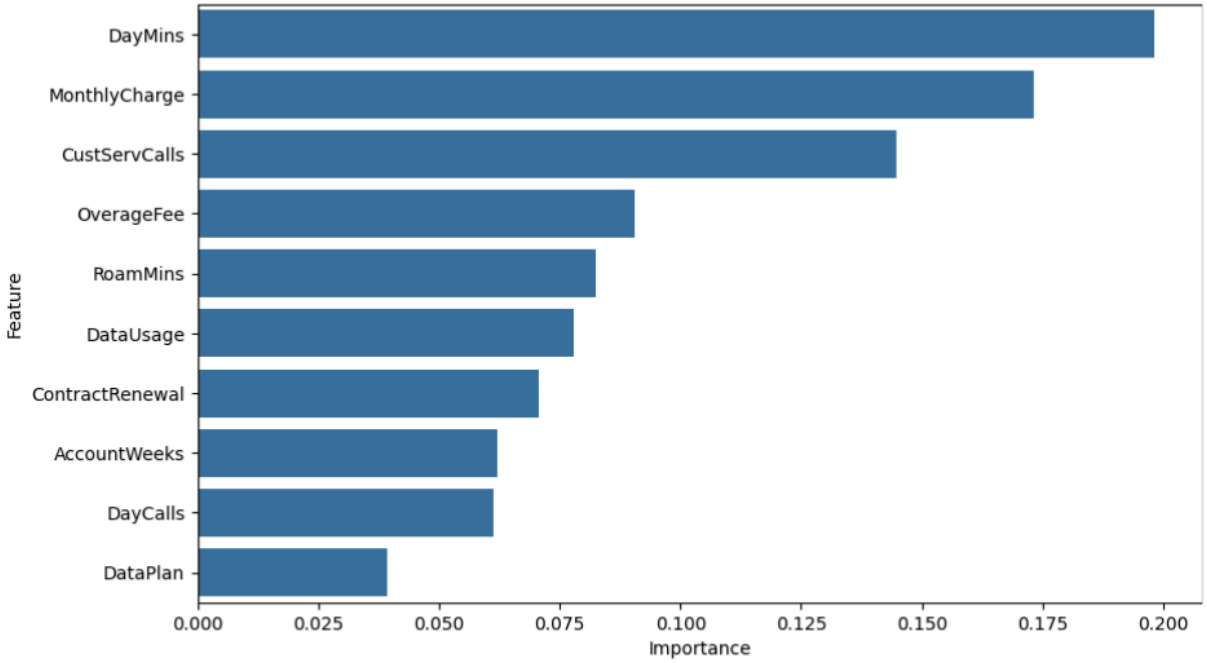
The Random Forest model will therefore be used for subsequent customer risk assessment, feature interpretation and high-risk customer identification.

8. Customer Risk and Segmentation

8.1 Feature Importance

The Random Forest feature-importance analysis identified the variables contributing most strongly to the model's predictive decisions.

Figure 10: Random Forest Feature Importance



The most influential features were:

1. DayMins — 19.82%
2. MonthlyCharge — 17.32%
3. CustServCalls — 14.48%
4. OverageFee — 9.06%
5. RoamMins — 8.24%
6. DataUsage — 7.80%
7. ContractRenewal — 7.06%
8. AccountWeeks — 6.20%
9. DayCalls — 6.11%
10. DataPlan — 3.92%

DayMins, MonthlyCharge and CustServCalls collectively account for approximately 51.6% of the Random Forest's feature importance, making them the most prominent individual predictors in the model.

NB: Feature importance indicates how useful variables are for the model's predictions, however, it does not establish that the variables causally drive churn.

8.2 Logistic Regression Interpretation

Logistic Regression provides an alternative, more directly interpretable view of the relationships between the variables and churn.

Table 6: Logistic Regression Coefficients and Odds Ratios

Feature	Coefficient	Odds Ratio
CustServCalls	0.73	2.07
DayMins	0.47	1.60
MonthlyCharge	0.33	1.39
OverageFee	0.28	1.32
RoamMins	0.21	1.23
DayCalls	0.08	1.09
AccountWeeks	0.03	1.04
DataUsage	-0.01	0.99
ContractRenewal	-0.58	0.56
DataPlan	-0.67	0.51

The odds ratios allow the direction and relative magnitude of associations to be interpreted while controlling for the other variables in the model.

For example, DayMins had an odds ratio of approximately 1.60, indicating that an increase by one standard deviation in day time usage was associated with 60% higher estimated odds of churn holding other model variables constant.

This provides complementary evidence to the Random Forest feature-importance analysis.

8.3 Model Explainability

SHAP analysis was used to examine how individual features and combinations of features contributed to the Random Forest's predictions.

Table 7: SHAP Feature Interaction Importance

Feature 1	Feature 2	Interaction Importance
DataUsage	MonthlyCharge	0.024
DataPlan	MonthlyCharge	0.016
CustServCalls	MonthlyCharge	0.012

ContractRenewal	RoamMins	0.011
DayMins	MonthlyCharge	0.010
CustServCalls	DayMins	0.010
DayMins	OverageFee	0.007
DataUsage	DayMins	0.007
DataPlan	DayMins	0.005
ContractRenewal	MonthlyCharge	0.005

The analysis indicates that churn risk is influenced not only by individual customer characteristics but also by interactions between characteristics. This supports the use of a tree-based model capable of capturing non-linear relationships and feature interactions.

8.4 High-Risk Customer Groups

The final Random Forest model was used to calculate predicted churn probabilities for existing customers. Analysis of the highest-risk customers identified several notable characteristics.

Table 8: High-Risk Customer Groups

Customer group	**Customers**	**Average predicted churn probability**
No Contract Renewal	186	13.86%
High DayMins + No Data Plan	377	9.31%
2+ Customer Service Calls	1,186	4.19%

The results indicate that contract renewal status is the strongest actionable segment-level indicator among the groups examined. High daytime usage combined with the absence of a data plan also identifies a comparatively elevated-risk group.

Frequent customer-service contact is more prevalent among high-risk customers, but the broader 2+ Customer Service Calls group has a substantially lower average predicted risk. It should therefore be treated as a supporting risk indicator rather than a standalone high-risk segment.

The relationship between daytime usage and predicted churn probability shows that higher usage is generally associated with a greater concentration of high-risk predictions. However, substantial variation remains among customers with similar usage levels, demonstrating that churn risk is determined by multiple customer characteristics rather than DayMins alone.

The model identifies customer usage, charges, service interactions and contractual/product characteristics as important components of churn risk. More importantly, the analysis translates these model-level findings into identifiable customer groups, with customers without contract renewal and high-usage customers without a data plan exhibiting the highest average predicted churn probabilities.

9. Business Implications & Retention Opportunities

The predictive analysis provides a basis for moving from identifying churn risk to targeting appropriate retention interventions. The objective should be to focus resources on customers where intervention is most likely to protect the customer relationship and recurring revenue.

9.1 Proactive Contract Renewal

Customers without contract renewal represent the highest-risk segment identified, with an average predicted churn probability of **13.86%.**

Opportunity: Prioritise these customers for proactive engagement before the renewal stage, using personalised renewal offers or plan options based on their individual usage and risk profile.

9.2 Targeted Product and Plan Recommendations

Customers with high daytime usage and no data plan have an average predicted churn probability of **9.31%.**

Opportunity: Review these customers for potential product-plan mismatch and provide targeted data-plan or bundled-plan recommendations where appropriate. This could improve perceived value while potentially increasing product adoption.

9.3 Risk-Based Customer Service Recovery

High-risk customers averaged **1.93 customer-service calls**, compared with **1.40** among other customers. However, the broader group of customers with two or more calls had an average predicted churn probability of only **4.19%**.

Opportunity: Use repeated service interactions as a supporting risk signal, rather than automatically treating every customer with multiple calls as a retention target. Customers

with repeated interactions and elevated modelled churn risk can be prioritised for issue resolution, escalation and follow-up.

9.4 Prioritisation of Retention Resources

The Random Forest provides an individual predicted churn probability, allowing customers to be ranked according to estimated risk.

This enables a tiered retention approach:

- Highest-risk customers: proactive, personalised intervention
- Moderate-risk customers: targeted offers or engagement
- Lower-risk customers: standard customer engagement

This approach can help the business avoid applying costly retention incentives uniformly and instead concentrate resources on customers with the greatest potential revenue-loss risk.

9.5 Business Impact

The analysis suggests that churn reduction should focus on targeted interventions rather than blanket retention campaigns. Improving the renewal experience, aligning products with customer usage and resolving persistent service issues can address different aspects of the customer relationship while allowing retention resources to be allocated according to predicted risk.

10. Limitations & Next Steps

10.1 Limitations

Model evaluation

The models were evaluated using a single train-test split. While this provides an independent test set, cross-validation would provide a more robust estimate of performance across different samples.

Risk assessment population

The high-risk customer analysis was performed by applying the fitted model to the same customer dataset used for model development. As a result, the predicted churn probabilities and identified customer segments may be more optimistic than when scoring previously unseen customers. This approach was taken to practically apply the model and identify potentially actionable customer groups to prevent potential churn.

Feature interpretation

Feature importance, odds ratios and SHAP values identify variables associated with model predictions. They do not establish causal relationships. For example, higher daytime usage being associated with greater churn risk does not mean that reducing a customer's usage would necessarily prevent churn.

Classification threshold

The models were evaluated using the standard classification threshold of 0.5. This may not represent the optimal threshold for a real-world retention programme, where the relative costs of false positives and false negatives should be considered.

Business impact

The analysis identifies customers who are predicted to be at higher risk of churn, but it does not establish whether a particular retention intervention will actually prevent those customers from leaving. Intervention effectiveness would need to be tested separately.

10.2 Next Steps

If this analysis were progressed towards production use, the following steps would strengthen the solution:

1. Cross-validation and hyperparameter tuning to improve confidence in model performance.
2. Threshold optimisation based on the financial cost of missed churn versus unnecessary retention interventions.
3. Probability calibration to ensure predicted churn probabilities accurately reflect observed outcomes.
4. A/B testing of retention strategies to establish which interventions actually reduce churn.
5. Model monitoring and periodic retraining to account for changes in customer behaviour and churn patterns.
6. Integration with customer-management systems so that high-risk customers can be identified and appropriate interventions triggered operationally.

10.3 Moving from Prediction to Action

The next stage should extend beyond model deployment to the development of a risk-based retention framework. Rather than applying the same intervention to every customer identified as high risk, the business should combine predicted churn risk, customer characteristics, intervention cost and expected customer value to determine the appropriate customer, timing and retention strategy for intervention.

This approach would enable retention resources to be prioritised where they are most likely to deliver meaningful customer and commercial impact.

11. Conclusion

This analysis demonstrates that customer churn can be predicted with a strong degree of accuracy using customer usage, service interaction, charges, product and contract characteristics. Of the three models evaluated, Random Forest provided the strongest overall performance, achieving 92.50% accuracy, 82.19% precision, 61.86% recall, an F1-score of 70.59% and an ROC-AUC of 86.32%.

The analysis identified daytime usage, monthly charges, customer-service interactions, contract renewal and data-plan status as important indicators of churn risk. Further analysis of predicted risk highlighted customers without contract renewal and high-usage customers without a data plan as particularly relevant customer groups for targeted retention activity.

The findings support a shift from broad, reactive retention efforts towards a more targeted, data-driven approach. By combining predicted churn risk with customer characteristics and commercial considerations, the business can prioritise customers for appropriate interventions, improve the relevance of the customer experience and focus retention resources where they have the greatest potential to protect recurring revenue.

Overall, the analysis provides a practical foundation for using predictive analytics to identify churn risk, understand customer behaviour and inform targeted retention strategies.